Project: Terminal Sets - Group 1.2

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***Abstract*— This report investigates the role of terminal sets for short MPC horizons in the context of a spring-mass-damper system. We discretize the system and formulate the MPC problem, implementing both polytopic and ellipsoidal terminal sets. The performance of the controllers is analyzed based on various metrics.**

I. INTRODUCTION

Model Predictive Control (MPC) is a versatile strategy widely adopted in industrial applications for its ability to manage complex, multivariable systems with constraints. It operates by solving optimization problems iteratively at each time step, using system models to predict future behavior and compute optimal control inputs. MPC finds extensive use in sectors such as chemical engineering, robotics, and aerospace, where precise control and adherence to constraints are crucial for operational success.[1][2]

A key aspect of MPC is the design of terminal sets, which ensure system stability and feasibility. Terminal sets come in two main types: polytopic and ellipsoidal. Polytopic sets are characterized by linear inequalities and are computationally straightforward to handle. On the other hand, ellipsoidal sets offer a more compact representation of the state space but involve more complex optimization processes. This study focuses on investigating the roles and performance implications of polytopic and ellipsoidal terminal sets within MPC frameworks, specifically examining their impact on stability, constraint satisfaction, computational efficiency, and closed-loop control performance.[3]

II. SYSTEM MODELING

A. Continuous-Time Model

The continuous-time model of the spring-mass-damper system is given by:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{1}{m}(F_{\text{ext}} - d \cdot x_2 - c \cdot x_1) \end{aligned} \tag{1}$$

where x_1 denotes the position of the mass and x_2 its velocity.

B. Discretization

The system is discretized using the forward Euler method with $\Delta t = 0.1$ s:

$$x_{k+1} = Ax_k + Bu_k \tag{2}$$

with the matrices A and B determine based on the parameters m, c, d, and Δt as:

$$A = \begin{pmatrix} 1 & \Delta t \\ -\frac{c\Delta t}{m} & 1 - \frac{d\Delta t}{m} \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ \frac{\Delta t}{m} \end{pmatrix} \tag{3}$$

III. MPC PROBLEM FORMULATION

A. Objective Function

The quadratic cost function to be minimized is:

$$J = \sum_{k=0}^{N-1} x_k^T Q x_k + \sum_{k=0}^{N-1} u_k^T R u_k + x_N^T P x_N \tag{4}$$

with weighting matrices:

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad R = 1 \tag{5}$$

B. Constraints

State and input constraints are defined as:

$$-1 \leq x_1 \leq 1, \quad -1 \leq x_2 \leq 1, \quad -0.68 \leq F_{\text{ext}} \leq 0.68 \tag{6}$$

C. Determination of LQR gain K and weight matrix P

Linear Quadratic Regulator (LQR) is used to design a controller that minimizes a quadratic cost function. The LQR gain K and the cost-to-go matrix P are derived from the Riccati equation.

D. Riccati Differential Equation

The Discrete-time Algebraic Riccati equation (DARE) is solved to obtain P:

$$P = A_d^T P A_d - A_d^T P B_d (R + B_d^T P B_d)^{-1} B_d^T P A_d + Q \tag{7}$$

The LQR gain K is computed as:

$$K = (R + B_d^T P B_d)^{-1} B_d^T P A_d \tag{8}$$

-Section I-III written by Harshdeep Singh-

IV. CLOSED LOOP TRAJECTORY

A. Theory

The previously discretized and stabilized system can now be simulated using Model Predictive Control.
For a finite prediction horizon N the inputs and the resulting states for the next N time instances can be predicted using the model described in section II. This computation will be done repeatedly for every step of the simulation and results in an optimal control trajectory of the states if N is chosen large enough.

B. *Prediction Horizon of N = 2 and N = 10*

To analyze the effect of an increasing prediction horizon on the closed loop trajectory the system will be simulated for a horizon of N = 2 and N = 10. The results can be seen in the figures 1 and 2.
When looking at the results for a prediction horizon of N = 2 one can see that although the states strive towards a steady state around X = (0,0) the state constraints have been violated during the simulation. This constraint violation can be prevented by increasing the prediction horizon as shown for N = 10. By increasing the prediction horizon the optimal control input can be adjusted preemptively to fulfill the state constraints. This can also be seen when looking at the input trajectories shown in figure 5. Because of the short prediction horizon of N = 2 the control input trajectories are more abrupt and extreme than for a horizon of N = 10.

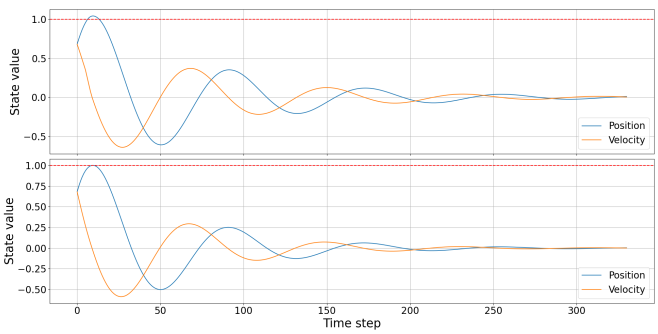

Fig. 1. State Trajectories for N = 2 (top) and N = 10 (bottom)

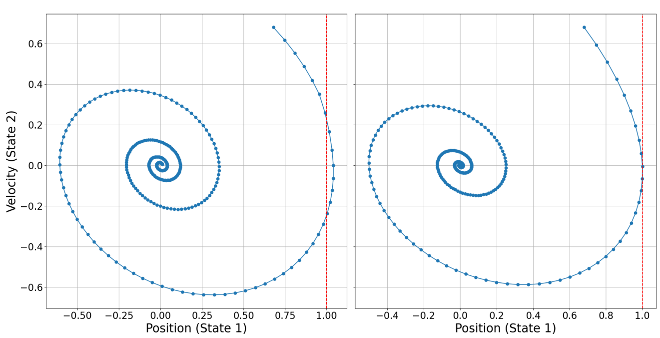

Fig. 2. State 2 vs. State 1 for N = 2 (left) and N = 10 (right)

Since the state predictions for N = 2 are closer to the current state, a violation of the constraints can not be anticipated fast enough and the closed loop reacts late. To not violate the constraints, the control input takes low values in order to steer the states away from their constraints. However, since the input values are also bounded the system response can not be strong enough to prevent a violation of the state constraints. This is implied by the jump of the control input values to their lower bound for N = 2. For a horizon of N = 10 the control input changes more steadily and in smaller steps during the simulation because the MPC looks further ahead and applies control inputs accordingly.

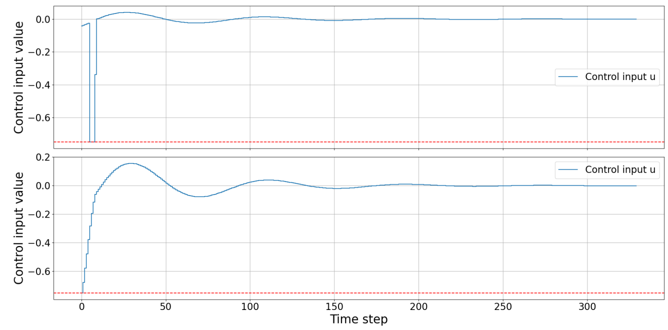

Fig. 3. Control Input Trajectories for N = 2 (top) and N = 10 (bottom)

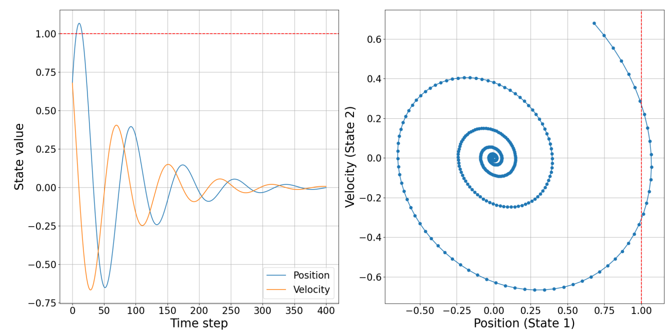

Fig. 4. State Trajectories (left) and State 2 vs. State 1 (right) with no Input

As previously mentioned the system strives for the steady state X = (0,0) for a horizon of N = 2. Both prediction horizons achieve that state in a similar simulation time, although figure 2 shows a horizon of N = 10 converging slightly faster. It might also be interesting to look at the behavior of the system if it is not controlled. This behavior can be seen in figure 4. One can see that although the system is not being controlled, the states still converge towards a steady state, meaning the system matrix A already stabilizes the system by itself. Compared to the applied MPC the simulation time needed has to be increased considerably to achieve the steady state if no control is applied. Furthermore the state constraints are being violated even stronger than with an MPC with a prediction horizon of N = 2 which shows that an implementation of an MPC is beneficial to the simulation of the system.
-Section IV written by Max Charles Heischmann-

V. Implementation of Polytopic Terminal Set

The aim is to enhance system stability and performance by incorporating a polytopic terminal set as a terminal constraint. The results demonstrate the effectiveness of this approach in maintaining the desired state trajectories and control inputs within predefined constraints. As we see in the system simulation with a short prediction horizon of 2 (N=2), the objective is to demonstrate how this approach can be applied to maintain the system states within a desired region

A. Definition and Importance

A polytopic terminal set is a convex polytope in the state space that serves as a target set for the system states at the end of a finite time horizon in MPC. Incorporating this set helps guarantee that the system states will remain within a specified region, thereby improving stability and performance. For a spring-mass-damper system, this can mean more precise control of oscillations and damping behavior.

B. Formulation

The terminal set is constructed iteratively by computing the maximum invariant set for the closed-loop system. The method for computing the maximum invariant set is as shown below[4]:

System dynamics: $x(k+1) = Ax(k) + Bu(k)$

State and input constraints:

$$\mathscr{X} := \{x \mid Fx \leq 1\},\ \mathscr{U} := \{u \mid Gu \leq 1\}$$

Choose a stabilizing feedback gain K for the system.

Apply $u(k) = Kx(k),\ \Rightarrow x_{k+1} = (A+BK)x(k)$

The maximal positively invariant set is described by

$$\Omega = \{x \mid F(A+BK)^i x \leq 1,\ GK(A+BK)^i x \leq 1,\ i = 0, \ldots, n_f\}$$

where n_f is the smallest positive integer such that

$$F(A+BK)^{n_f+1} x \leq 1,\ GK(A+BK)^{n_f+1} x \leq 1$$

holds for all x satisfying

$$F(A+BK)^i x \leq 1,\ GK(A+BK)^i x \leq 1,\ i = 0, \ldots, n_f$$

C. Results

After implementing the terminal sets, we can see the polytopic terminal set in Fig.5. The simulation results are as follows:

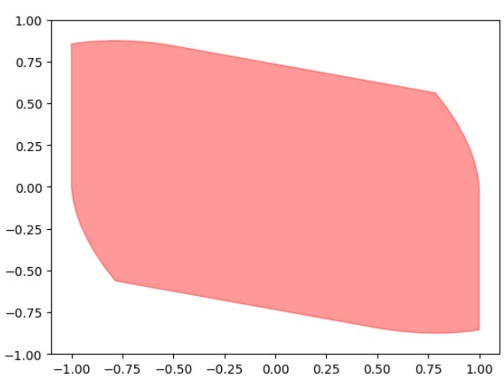

Fig. 5. Polytopic terminal set for N=2

D. State Trajectories and State Space Analysis

The position and velocity trajectories remain within the specified bounds, demonstrating the effectiveness of the polytopic terminal set in maintaining feasibility.Fig.6

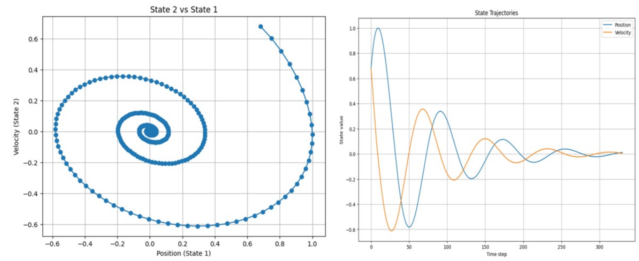

Fig. 6. State Trajectories for N=2 in polytopic terminal

E. Control Inputs

The control inputs are well-regulated within the constraints, showing smooth and feasible control actions.Fig.7

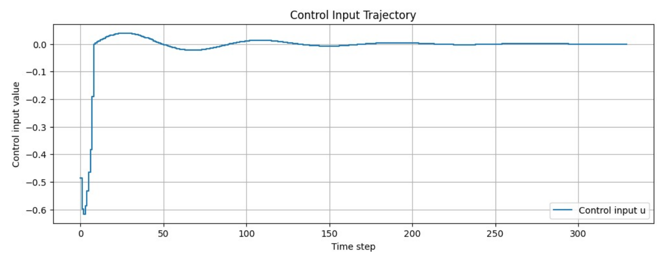

Fig. 7. Control inputs Trajectories for N=2 in polytopic terminal set

-Section V and VII written by Nasim vosoughi osgouei

VI. Ellipsoidal Invariant Sets

A. Controlled positively invariant set

A set $\mathscr{X} \subseteq \mathbb{R}^{n_x}$ is controlled positively invariant (CPI) for the dynamics of (1) and constraints (6) if, for all $x \in \mathscr{X}$, there exists $u \in \mathbb{R}^{n_u}$ such that $Fx + Gu \leq 1$ and $Ax + Bu \in \mathscr{X}$[3].

Ellipsoids are very popular as candidate invariant sets. An ellipsoidal set can be always represented as follows[2]:

$$S = \{x \in \mathbb{R}^n : x^T P x \leq 1\} \tag{9}$$

where P is a symmetric positive-definite matrix.

B. Formulation

The ellipsoidal set defined by $E_z = \{z : z^T P_z z \leq 1\}$ can be determine by solving the semi definite program for the positively invariant ellipsoidal set given by [3][1]:

$$\max_{(S,H)} \log\det(S_{xx}) \tag{10}$$

Subject to:

$$\begin{bmatrix} S & \Psi S \\ S\Psi^T & S \end{bmatrix} \succeq 0 \quad \text{which is an LMI in } S \tag{11}$$

$$\begin{bmatrix} H & [F+GKGE]S \\ S[F+GKGE]^T & S \end{bmatrix} \succeq 0 \quad \text{LMI in } S \text{ and } H \tag{12}$$

$$e_i^T H e_i \leq 1, \quad i = 1, \ldots, n_c \tag{13}$$

where e_i is the ith column of the identity matrix.

Ψ is given by:

$$\Psi = \begin{bmatrix} A+BK & BE \\ 0 & M \end{bmatrix} \tag{14}$$

M and E are given by

$$M = \begin{bmatrix} 0 & I_{n_u} & 0 & \cdots & 0 \\ 0 & 0 & I_{n_u} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & I_{n_u} \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} E = \begin{bmatrix} I_{n_u} & 0 & \cdots & 0 \end{bmatrix}$$

C. Results

After implementing the ellipsoidal terminal set algorithm, we can see the terminal set ontop of polytopic terminal set Fig.8. The simulation results are as follows:

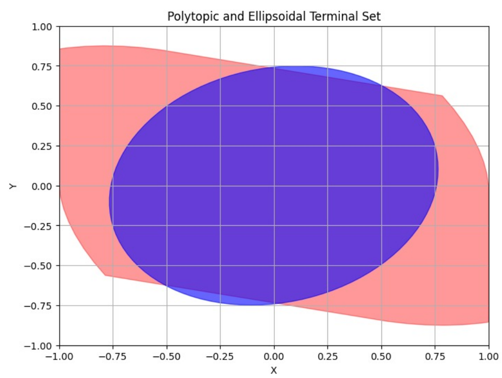

Fig. 8. Ellipsoidal and polytopic terminal set for N=2

D. State Trajectorie

The position and velocity trajectories remain within the specified bounds, demonstrating the effectiveness of the ellipsoidal terminal set in maintaining feasibility.Fig.9

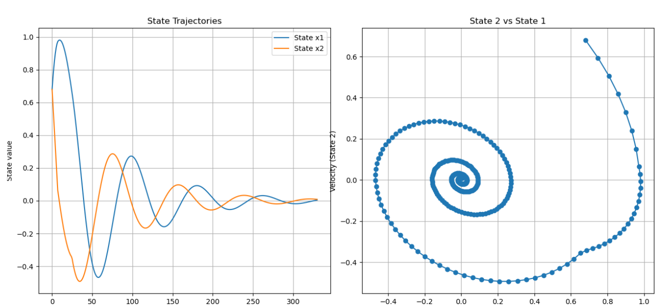

Fig. 9. State Trajectories for N=2 in ellipsoidal termina set

E. Control Inputs

The control inputs are well-regulated within the constraints, showing smooth and feasible control actions.Fig.10

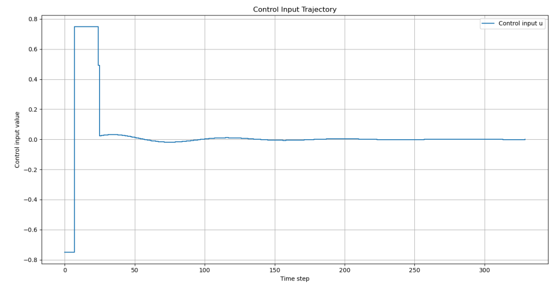

Fig. 10. Control inputs Trajectories for N=2 in ellipsoidal terminal set

F. Performance of different Methods

The table below show the Performance metric of MPC horizon length N=10, and MPC horizon length N=2 with terminal and ellipsoidal set for the mass spring damper system.

TABLE I
PERFORMANCE COMPARISON

Horizon	***Number of constraints***	***Computational time***	***Value cost***
N=10	20	2.39s	1.389e-07
N=2 Polytopic	604	16.65s	0.00042
N=2 Ellipsoidal	21	2.9s	0.00016

Below show the plot of the value function

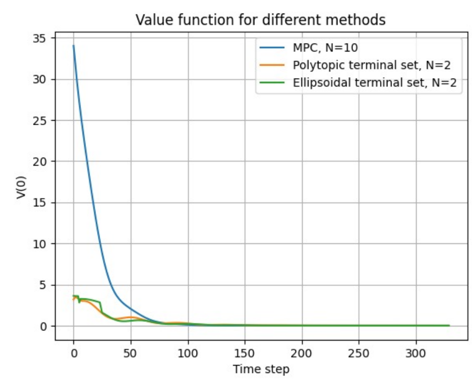

Fig. 11. Optimal value function

VII. Conclusion

The value function for all three methods decreases over time, eventually converging to a reasonable steady-state value.Fig.11 The polytopic terminal set with N=2 has a significantly higher computational time due to a large number of inequality constraints. This method, while computationally intensive, provides robust performance within its constraints. The ellipsoidal terminal set with N=2 balances computational time and performance, making it a suitable choice when both factors need to be optimized. (Table.I) For our case of study, considering all the results show that Ellipsoidal terminal set can be a good choice, also MPC with N=10 can be a reasonable choice too.

-Section VI-VII written by Desmond Fotock Fomelack

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